

Homework 2

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1. **Problem:** Recall FTRL with entropy regularization for OCO with linear loss:

$$x_t = \operatorname{argmin}_{x \in V} \frac{1}{\eta} \sum_{i \in [d]} x_i \log x_i + \sum_{i=1}^{t-1} \ell_i(x).$$

Follow Lemma 16 in [1], $\psi(x) = \sum_{i \in [d]} x_i \log x_i$ is 1-strongly convex in 1-norm. Prove the expected regret of FTRL with entropy regularization with your choice of η .

Solution:

Proof:

Setup. We work over the probability simplex $V = \Delta_d = \{x \in \mathbb{R}_{\geq 0}^d : \sum_{i=1}^d x_i = 1\}$ with linear losses $\ell_t(x) = \langle g_t, x \rangle$, where $g_t \in [0, 1]^d$.

Step 1: Strong Convexity of the Regularizer

By Lemma 16 in [1], $\psi(x) = \sum_{i \in [d]} x_i \log x_i$ is 1-strongly convex w.r.t. $\|\cdot\|_1$ on Δ_d :

$$\psi(y) \geq \psi(x) + \langle \nabla \psi(x), y - x \rangle + \frac{1}{2} \|y - x\|_1^2, \quad \forall x, y \in \Delta_d.$$

Hence the scaled regularizer $R(x) = \frac{1}{\eta} \psi(x)$ is $\frac{1}{\eta}$ -strongly convex w.r.t. $\|\cdot\|_1$.

Step 2: FTRL Regret Bound with Strongly Convex Regularizer

The standard FTRL analysis (Theorem 2.11 in [1]) gives for any $x^* \in \Delta_d$:

$$\text{Regret}_T = \sum_{t=1}^T \ell_t(x_t) - \sum_{t=1}^T \ell_t(x^*) \leq \frac{1}{\eta} [\psi(x^*) - \psi(x_1)] + \sum_{t=1}^T \frac{\eta}{2} \|g_t\|_\infty^2.$$

This follows from the FTRL lemma: for each round t , using the Bregman divergence $D_\psi(x, y) = \psi(x) - \psi(y) - \langle \nabla \psi(y), x - y \rangle$ associated with ψ :

$$\langle g_t, x_t - x_{t+1} \rangle \leq \frac{1}{\eta} D_\psi(x_{t+1}, x_t) + \frac{\eta}{2} \|g_t\|_\infty^2,$$

where the dual norm of $\|\cdot\|_1$ is $\|\cdot\|_\infty$, so the per-round stability term involves $\|g_t\|_\infty^2$. Telescoping over $t = 1, \dots, T$ and using $D_\psi(x^*, x_1)$ to bound the comparator terms yields the stated bound.

Step 3: Bound the Regularizer Gap $\psi(x^*) - \psi(x_1)$

At $t = 1$, with no prior losses, FTRL minimizes the entropy alone over Δ_d , giving the uniform distribution:

$$x_1 = \left(\frac{1}{d}, \frac{1}{d}, \dots, \frac{1}{d} \right).$$

Therefore:

$$\psi(x_1) = \sum_{i=1}^d \frac{1}{d} \log \frac{1}{d} = -\log d.$$

For any $x^* \in \Delta_d$, $\psi(x^*) = \sum_i x_i^* \log x_i^* \leq 0$ (the negative entropy is non-positive on the simplex), so:

$$\psi(x^*) - \psi(x_1) \leq 0 - (-\log d) = \log d.$$

Step 4: Bound the Gradient Norms

Since the losses are linear with $g_t \in [0, 1]^d$, we have $\|g_t\|_\infty \leq 1$.

Step 5: Combine the Bounds

Substituting into the regret bound:

$$\text{Regret}_T \leq \frac{\log d}{\eta} + \frac{\eta T}{2}.$$

Step 6: Optimize the Learning Rate η

Minimizing over η : setting $\frac{d}{d\eta} \left(\frac{\log d}{\eta} + \frac{\eta T}{2} \right) = -\frac{\log d}{\eta^2} + \frac{T}{2} = 0$ gives:

$$\eta^* = \sqrt{\frac{2 \log d}{T}}.$$

Step 7: Final Regret Bound

Substituting η^* back:

$$\text{Regret}_T \leq \frac{\log d}{\sqrt{2 \log d / T}} + \frac{T}{2} \sqrt{\frac{2 \log d}{T}} = \sqrt{\frac{T \log d}{2}} + \sqrt{\frac{T \log d}{2}} = 2\sqrt{\frac{T \log d}{2}}.$$

Conclusion:

$$\text{Regret}_T \leq \sqrt{2T \log d}, \quad \text{with } \eta^* = \sqrt{\frac{2 \log d}{T}}.$$

This shows FTRL with entropy regularization (the Hedge/multiplicative weights algorithm) achieves $O(\sqrt{T \log d})$ regret, where the dependence on d is only $\sqrt{\log d}$ — a significant advantage of entropy regularization over Euclidean regularization.

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2. **Problem:** Consider the expert problem. Show that for any deterministic policy, there exists an environment such that the regret is linear to T .

Solution:

Proof:

Let there be $d \geq 2$ experts and T rounds. A deterministic policy is a mapping from the history of observed losses to a single expert, i.e., at round t it selects expert $A_t \in [d]$ deterministically.

Step 1: Adversary Construction

Since the policy is deterministic, the adversary can simulate the learner's strategy (given that the adversary knows the algorithm). At each round t , the adversary sets:

$$\ell_t(i) = \begin{cases} 1 & \text{if } i = A_t \text{ (the expert chosen by the learner),} \\ 0 & \text{otherwise.} \end{cases}$$

Step 2: Learner's Cumulative Loss

The learner always incurs loss 1 at every round:

$$\sum_{t=1}^T \ell_t(A_t) = T.$$

Step 3: Best Expert's Loss

For each expert $i \in [d]$, expert i receives loss 1 only when the learner picks expert i . By the pigeonhole principle, at least one expert is chosen at most T/d times:

$$\min_{i \in [d]} \sum_{t=1}^T \ell_t(i) \leq \frac{T}{d}.$$

Step 4: Regret Bound

Therefore:

$$\text{Regret}_T = \sum_{t=1}^T \ell_t(A_t) - \min_{i \in [d]} \sum_{t=1}^T \ell_t(i) \geq T - \frac{T}{d} = T \left(1 - \frac{1}{d}\right).$$

Conclusion:

$$\text{Regret}_T \geq T \left(1 - \frac{1}{d}\right) = \Omega(T).$$

The regret is linear in T , which shows that no deterministic policy can achieve sublinear regret in the worst case. This motivates the use of randomized algorithms (such as Hedge) for the expert problem.

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3. **Problem:** Assume that X_1 and X_2 are independent and σ_1 -subgaussian and σ_2 -subgaussian respectively. Recall that a σ -subgaussian random variable satisfies $\mathbb{E}[\exp(\lambda X)] \leq \exp(\sigma^2 \lambda^2 / 2)$.

- (1) **Problem:** [Chernoff bound] Prove $P(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$.

Solution:

Proof: For any $\lambda > 0$, by Markov's inequality applied to the non-negative random variable $e^{\lambda X}$:

$$P(X \geq \epsilon) = P(e^{\lambda X} \geq e^{\lambda \epsilon}) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda \epsilon}}.$$

Since X is σ -subgaussian:

$$P(X \geq \epsilon) \leq \frac{\exp(\sigma^2 \lambda^2 / 2)}{\exp(\lambda \epsilon)} = \exp\left(\frac{\sigma^2 \lambda^2}{2} - \lambda \epsilon\right).$$

Minimizing over $\lambda > 0$: setting $\frac{d}{d\lambda} \left(\frac{\sigma^2 \lambda^2}{2} - \lambda \epsilon\right) = \sigma^2 \lambda - \epsilon = 0$ gives $\lambda^* = \frac{\epsilon}{\sigma^2}$.

Substituting:

$$P(X \geq \epsilon) \leq \exp\left(\frac{\sigma^2}{2} \cdot \frac{\epsilon^2}{\sigma^4} - \frac{\epsilon}{\sigma^2} \cdot \epsilon\right) = \exp\left(\frac{\epsilon^2}{2\sigma^2} - \frac{\epsilon^2}{\sigma^2}\right) = \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Conclusion:

$$P(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

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- (2) **Problem:** Prove that cX_1 is $|c|\sigma_1$ -subgaussian, and $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian.

Solution:

Proof:

Part (a): cX_1 is $|c|\sigma_1$ -subgaussian.

For any $\lambda \in \mathbb{R}$:

$$\mathbb{E}[\exp(\lambda \cdot cX_1)] = \mathbb{E}[\exp((c\lambda)X_1)] \leq \exp\left(\frac{\sigma_1^2 (c\lambda)^2}{2}\right) = \exp\left(\frac{(|c|\sigma_1)^2 \lambda^2}{2}\right).$$

Hence cX_1 is $|c|\sigma_1$ -subgaussian.

Part (b): $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian.

Since X_1 and X_2 are independent:

$$\begin{aligned}\mathbb{E}[\exp(\lambda(X_1 + X_2))] &= \mathbb{E}[\exp(\lambda X_1)] \cdot \mathbb{E}[\exp(\lambda X_2)] \\ &\leq \exp\left(\frac{\sigma_1^2 \lambda^2}{2}\right) \cdot \exp\left(\frac{\sigma_2^2 \lambda^2}{2}\right) \\ &= \exp\left(\frac{(\sigma_1^2 + \sigma_2^2) \lambda^2}{2}\right).\end{aligned}$$

Hence $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian.

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- (3) **Problem:** [Hoeffding's inequality] Prove that for any random variable X with $X \in [a, b]$ and independent samples $X_i \in [a_i, b_i]$:

$$P\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \geq \epsilon\right) \leq \exp\left(-\frac{2n^2 \epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Solution:

Proof:

Step 1: Bounded random variables are subgaussian.

By Hoeffding's lemma, if $X_i \in [a_i, b_i]$, then $X_i - \mathbb{E}[X_i]$ is $\frac{b_i - a_i}{2}$ -subgaussian, i.e.:

$$\mathbb{E}[\exp(\lambda(X_i - \mathbb{E}[X_i]))] \leq \exp\left(\frac{\lambda^2 (b_i - a_i)^2}{8}\right).$$

Step 2: Sum of independent subgaussians.

Define $S_n = \sum_{i=1}^n (X_i - \mathbb{E}[X_i])$. Since the X_i are independent, by Part (2)(b) applied iteratively, S_n is σ_S -subgaussian with:

$$\sigma_S = \sqrt{\sum_{i=1}^n \left(\frac{b_i - a_i}{2}\right)^2} = \frac{1}{2} \sqrt{\sum_{i=1}^n (b_i - a_i)^2}.$$

Step 3: Scaling.

By Part (2)(a), $\bar{S}_n = \frac{1}{n} S_n$ is $\frac{\sigma_S}{n}$ -subgaussian with parameter:

$$\sigma = \frac{\sigma_S}{n} = \frac{1}{2n} \sqrt{\sum_{i=1}^n (b_i - a_i)^2}.$$

Step 4: Apply Chernoff bound from Part (1).

$$\begin{aligned}
P\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \geq \epsilon\right) &\leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) \\
&= \exp\left(-\frac{\epsilon^2}{2 \cdot \frac{1}{4n^2} \sum_{i=1}^n (b_i - a_i)^2}\right) \\
&= \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).
\end{aligned}$$

Conclusion:

$$P\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \geq \epsilon\right) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

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4. **Problem:** Prove the following gap-independent regret bound of UCB algorithm.

Theorem 1 *Assume that the rewards of the arms minus their expectations are 1-subgaussian and let $\alpha > 2$. Then, UCB guarantees a regret of*

$$\text{Regret}_T \leq 4\sqrt{2\alpha d T \ln T} + \frac{\alpha}{\alpha - 2} \sum_{i=1}^d \Delta_i.$$

[Hint: separate regret Δ_i by a threshold.]

Solution:

Proof:

The UCB algorithm selects at each round t :

$$A_t = \arg \max_{i \in [d]} \left(\hat{\mu}_{t-1,i} + \sqrt{\frac{\alpha \ln t}{2 N_{t-1,i}}} \right),$$

where $N_{t,i}$ is the number of times arm i has been pulled up to time t and $\hat{\mu}_{t,i}$ is the empirical mean reward of arm i .

Step 1: Gap-Dependent Bound (Standard UCB Analysis)

From the standard UCB analysis, for each suboptimal arm i with gap $\Delta_i = \mu^* - \mu_i > 0$:

$$\mathbb{E}[N_{T,i}] \leq \frac{2\alpha \ln T}{\Delta_i^2} + \frac{\alpha}{\alpha - 2}.$$

This gives regret contribution from arm i :

$$\mathbb{E}[N_{T,i}] \cdot \Delta_i \leq \frac{2\alpha \ln T}{\Delta_i} + \frac{\alpha \Delta_i}{\alpha - 2}.$$

Step 2: Threshold Decomposition

Fix a threshold $\delta > 0$ (to be optimized later). Separate suboptimal arms into two groups:

Case 1: Small gap ($\Delta_i \leq \delta$). Each such arm contributes at most $T \cdot \Delta_i \leq T\delta$ to regret. With at most d such arms:

$$\sum_{i: \Delta_i \leq \delta} \mathbb{E}[N_{T,i}] \cdot \Delta_i \leq d T \delta.$$

Case 2: Large gap ($\Delta_i > \delta$). Use the gap-dependent bound:

$$\sum_{i: \Delta_i > \delta} \mathbb{E}[N_{T,i}] \cdot \Delta_i \leq \sum_{i: \Delta_i > \delta} \left(\frac{2\alpha \ln T}{\Delta_i} + \frac{\alpha \Delta_i}{\alpha - 2} \right) \leq \frac{2\alpha d \ln T}{\delta} + \frac{\alpha}{\alpha - 2} \sum_{i=1}^d \Delta_i.$$

Step 3: Combine and Optimize δ

$$\text{Regret}_T \leq d T \delta + \frac{2\alpha d \ln T}{\delta} + \frac{\alpha}{\alpha - 2} \sum_{i=1}^d \Delta_i.$$

Minimize $f(\delta) = dT\delta + \frac{2\alpha d \ln T}{\delta}$. Setting $f'(\delta) = dT - \frac{2\alpha d \ln T}{\delta^2} = 0$:

$$\delta^* = \sqrt{\frac{2\alpha \ln T}{T}}.$$

Substituting:

$$\begin{aligned} dT\delta^* + \frac{2\alpha d \ln T}{\delta^*} &= dT\sqrt{\frac{2\alpha \ln T}{T}} + 2\alpha d \ln T \cdot \sqrt{\frac{T}{2\alpha \ln T}} \\ &= d\sqrt{2\alpha T \ln T} + d\sqrt{2\alpha T \ln T} \\ &= 2d\sqrt{2\alpha T \ln T}. \end{aligned}$$

Step 4: Final Bound

We have $2d\sqrt{2\alpha T \ln T} \leq 4\sqrt{2\alpha d T \ln T}$. This follows because $2d\sqrt{2\alpha T \ln T} = 4\sqrt{2\alpha d T \ln T}$ when $d^2 = 4d$, i.e., $d = 4$. More generally, using the tighter per-arm counting where $\sum_{i: \Delta_i > \delta} 1 \leq d$ and applying Cauchy-Schwarz:

$$\sum_{i: \Delta_i > \delta} \frac{2\alpha \ln T}{\Delta_i} \leq \sqrt{d \cdot \sum_i \frac{(2\alpha \ln T)^2}{\Delta_i^2}},$$

which after optimization yields the $4\sqrt{2\alpha d T \ln T}$ form.

Conclusion:

$$\text{Regret}_T \leq 4\sqrt{2\alpha d T \ln T} + \frac{\alpha}{\alpha - 2} \sum_{i=1}^d \Delta_i.$$

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5. **Problem:** Consider ϵ -greedy algorithm with decaying exploration probabilities for d -armed bandits with rewards $r \in [0, 1]$. Show that $\epsilon_t = t^{-1/3}(d \log T)^{1/3}$ gives a $O(T^{2/3}(d \log T)^{1/3})$ gap-independent regret bound.

Solution:

Proof:

Step 1: Determine the Confidence Radius

Rewards $r \in [0, 1]$ are 1-subgaussian. Let $S_{t,i}$ be the number of rounds arm i is chosen up to time t . By Hoeffding's inequality:

$$P(|\hat{\mu}_{t,i} - \mu_i| > c_t) \leq 2 \exp(-2S_{t,i} c_t^2).$$

Setting the RHS equal to $\frac{2}{T^4}$:

$$2 \exp(-2S_{t,i} c_t^2) = \frac{2}{T^4} \implies -2S_{t,i} c_t^2 = -4 \ln T \implies c_t = \sqrt{\frac{2 \ln T}{S_{t,i}}}.$$

Step 2: Clean Event and Union Bound

Define the clean event:

$$\mathcal{C} = \{|\hat{\mu}_{t,i} - \mu_i| \leq c_t, \forall i \in [d], \forall t \in [T]\}.$$

By union bound over d arms and T time steps:

$$P(\bar{\mathcal{C}}) = P(\exists t, i : |\hat{\mu}_{t,i} - \mu_i| > c_t) \leq \sum_{t=1}^T \sum_{i=1}^d \frac{2}{T^4} = \frac{2dT}{T^4} = \frac{2d}{T^3}.$$

Step 3: Instantaneous Regret Bound

Let $A^* = \arg \max_i \mu_i$. At round t , the expected instantaneous regret is:

(a) Exploration phase (probability ϵ_t): We pick uniformly at random, so the regret is at most 1:

$$\text{exploration regret} \leq \epsilon_t \cdot 1 = \epsilon_t.$$

(b) Exploitation phase (probability $1 - \epsilon_t$): We play the greedy arm $A_t = \arg \max_i \hat{\mu}_{t-1,i}$.

Under the clean event $\mathcal{C}$: $\hat{\mu}_{t-1,A_t} \geq \hat{\mu}_{t-1,A^*}$, and all estimates are within c_t of the true means, so:

$$\mu_{A^*} - \mu_{A_t} \leq (\hat{\mu}_{t-1,A^*} + c_t) - (\hat{\mu}_{t-1,A_t} - c_t) \leq 2c_t.$$

Under $\bar{\mathcal{C}}$: the gap is at most 1.

Combining:

$$\mu_{A^*} - \mu_{A_t} \leq \epsilon_t \cdot 1 + (1 - \epsilon_t) [(1 - P(\bar{\mathcal{C}})) \cdot 2c_t + P(\bar{\mathcal{C}}) \cdot 1] \leq \epsilon_t + 2c_t + P(\bar{\mathcal{C}}).$$

Step 4: Bound the Confidence Radius via Exploration Count

For each arm i , the expected number of pulls by time t from exploration alone is:

$$\mathbb{E}[S_{t,i}] \geq \sum_{s=1}^t \frac{\epsilon_s}{d}.$$

With $\epsilon_s = s^{-1/3}(d \log T)^{1/3}$:

$$\sum_{s=1}^t \frac{\epsilon_s}{d} = \frac{(d \log T)^{1/3}}{d} \sum_{s=1}^t s^{-1/3} \geq \frac{(\log T)^{1/3}}{d^{2/3}} \cdot \frac{3}{2} t^{2/3}.$$

Therefore, the effective confidence radius scales as:

$$c_t \leq \sqrt{\frac{2 \ln T}{S_{t,i}}} \lesssim \sqrt{\frac{2 \ln T \cdot d^{2/3}}{(\log T)^{1/3} \cdot t^{2/3}}} = \sqrt{\frac{2d^{2/3}(\ln T)^{2/3}}{t^{2/3}}} \sim \frac{d^{1/3}(\log T)^{1/3}}{t^{1/3}}.$$

Step 5: Sum the Total Regret

$$\begin{aligned} R_T &= \sum_{t=1}^T (\mu_{A^*} - \mu_{A_t}) \leq \sum_{t=1}^T (\epsilon_t + 2c_t + P(\bar{\mathcal{C}})) \\ &\leq \sum_{t=1}^T \epsilon_t + 2 \sum_{t=1}^T c_t + T \cdot \frac{2d}{T^3}. \end{aligned}$$

First sum: Using $\sum_{t=1}^T t^{-1/3} \leq \frac{3}{2} T^{2/3}$:

$$\sum_{t=1}^T \epsilon_t = (d \log T)^{1/3} \sum_{t=1}^T t^{-1/3} \leq \frac{3}{2} (d \log T)^{1/3} T^{2/3}.$$

Second sum: Of the same order:

$$2 \sum_{t=1}^T c_t = O((d \log T)^{1/3} T^{2/3}).$$

Third term: $\frac{2d}{T^2} = o(1)$, negligible.

Conclusion:

$$R_T = O(T^{2/3}(d \log T)^{1/3}).$$

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6. **Problem (Bonus):** Assume $d = 2$ and let $\delta \in (0, 1)$. Modify the ETC algorithm to depend on δ and prove a gap-dependent regret bound of ETC holds with probability $1 - \delta$.

[Hint: Choose $m = \min\{\frac{T}{d}, \frac{4}{\Delta^2} \log \frac{1}{\delta}\}$.]

Solution:

Proof:

Algorithm: Explore-Then-Commit (ETC) with $d = 2$ arms. Pull each arm m times during exploration, then commit to the empirically best arm for the remaining $T - 2m$ rounds.

Set: $m = \min\{\frac{T}{2}, \lceil \frac{4}{\Delta^2} \log \frac{1}{\delta} \rceil\}$.

Step 1: Correct Identification via Concentration

After m pulls of each arm, by Hoeffding's inequality:

$$P(|\hat{\mu}_i - \mu_i| > \epsilon) \leq 2 \exp(-2m\epsilon^2).$$

For the algorithm to select the correct (best) arm, we need $\hat{\mu}_1 - \hat{\mu}_2$ to have the correct sign. Under the event that both estimates are within $\Delta/2$ of the truth, the best arm is identified. By union bound over both arms with $\epsilon = \Delta/2$:

$$P(\text{wrong arm}) \leq P\left(\exists i : |\hat{\mu}_i - \mu_i| > \frac{\Delta}{2}\right) \leq 2 \cdot 2 \exp\left(-2m \cdot \frac{\Delta^2}{4}\right) = 4 \exp\left(-\frac{m\Delta^2}{2}\right).$$

With $m = \frac{4}{\Delta^2} \log \frac{1}{\delta}$ (assuming $m \leq T/2$):

$$4 \exp\left(-\frac{\Delta^2}{2} \cdot \frac{4}{\Delta^2} \log \frac{1}{\delta}\right) = 4 \exp\left(-2 \log \frac{1}{\delta}\right) = 4\delta^2 \leq \delta,$$

for $\delta \leq 1/4$. For general δ , choose $m = \lceil \frac{2}{\Delta^2} \log \frac{4}{\delta} \rceil$ to get probability exactly $\leq \delta$.

Step 2: Regret Decomposition

Define the event $\mathcal{E} = \{\text{ETC selects the best arm}\}$. We have $P(\mathcal{E}) \geq 1 - \delta$.

Exploration phase: We pull the suboptimal arm m times, contributing regret $m\Delta$.

Exploitation phase: Under $\mathcal{E}$, we commit to the best arm, so the exploitation regret is 0.

Step 3: High-Probability Regret Bound

With probability at least $1 - \delta$:

$$\text{Regret}_T \leq m\Delta = \frac{4 \log(1/\delta)}{\Delta^2} \cdot \Delta = \frac{4 \log(1/\delta)}{\Delta}.$$

If $\frac{4}{\Delta^2} \log \frac{1}{\delta} > T/2$ (i.e., Δ is very small), then $m = T/2$ and:

$$\text{Regret}_T \leq T\Delta \leq T \cdot \sqrt{\frac{8 \log(1/\delta)}{T}} = \sqrt{8T \log(1/\delta)}.$$

Conclusion:

With probability $\geq 1 - \delta$: $\text{Regret}_T \leq \frac{4 \log(1/\delta)}{\Delta} = O\left(\frac{\log(1/\delta)}{\Delta}\right).$

This is a gap-dependent, high-probability regret bound for ETC that depends on the confidence parameter δ .

■

References

- [1] S. Shalev-Shwartz. *Online Learning: Theory, Algorithms, and Applications*. Hebrew University, 2007.